Channel, Pulse Compression and Tracking

Prof. Dr. Ali Kara

Objective

learning channel effects in radar operation, and understand pulse compression and tracking

Learning Outcomes

At the end of the lesson, you should be able to

1. describe some major channel effects commonly encountered in radar/EW operations
2. understand pulse compression concept in radar/EW
3. describe and compare three types of trackers, and discuss their operational constraints

Presentation Plan

- Channel effects in radar/EW operation
- Pulse compression
- Tracking radar
- Tracker types

Radar Range: more realistic

Channel Induced losses would include

- Clutter effects
- Lobing
- Radar horizon
- Ducting

Radar *range* $R_{max}^4 = \dfrac{P_t G A_e \sigma}{(4\pi)^2 k T_0 B F_n (S/N)_{min}}$

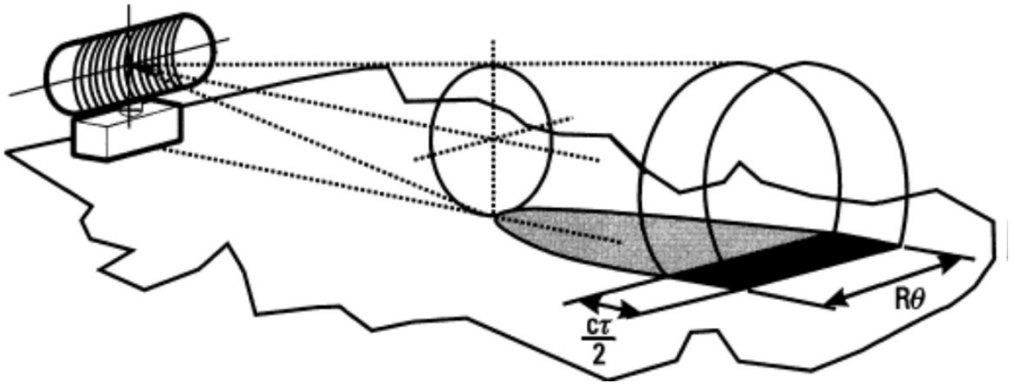
$$R_{max} = \bar{R}_{max} - Channel\ Induced\ Losses(F_i)$$

Clutter [Neri]

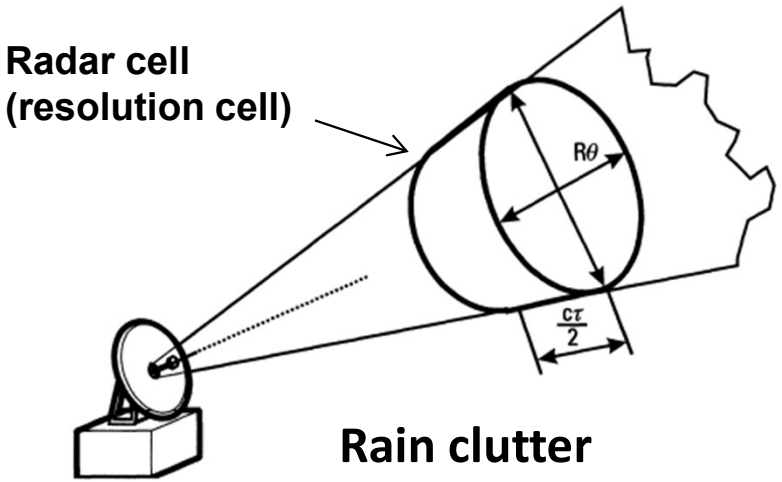
Clutter: unwanted echoes from the natural environment

- •Ground (ground reflectivity in the radar cell, frequency, low grazing angle case !)
- •Sea (reflectivity of the sea, frequency, angle)
- •Rain (volumetric, rainfall inside the radar cell)

Sea State SS	Wind State $K_s = SS + 1$	Wave height (rms) σ_h
1	2	0.003m
3	4	0.2m
5	6	0.7m
7	8	1.7m



Sea clutter



Rain clutter

[Skolnik]

Sea State	Meters	Descriptive Term
0	0	Calm, glassy
1	0–0.1	Calm, rippled
2	0.1–0.5	Smooth, wavelets
3	0.6–1.2	Slight
4	1.2–2.4	Moderate
5	2.4–4.0	Rough
6	4.0–6.0	Very rough
7	6.0–9.0	High
8	9.0–14	Very high
9	over 14	Phenomenal

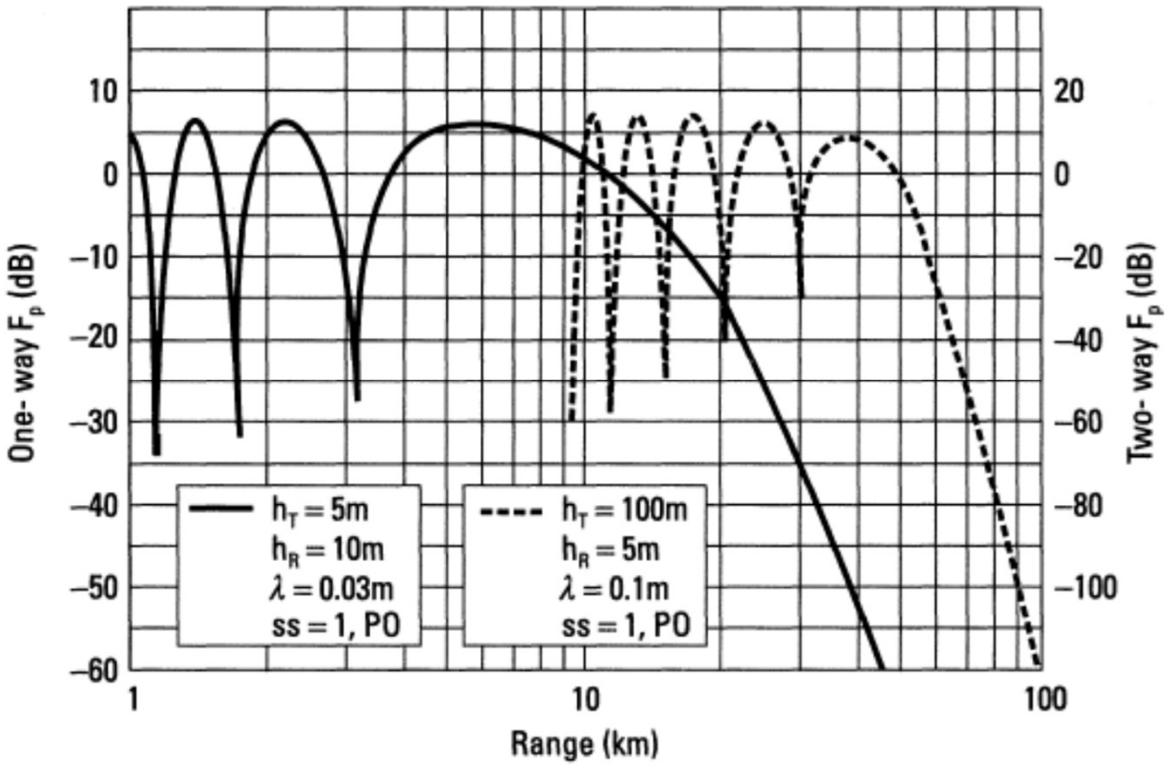
[Barton]

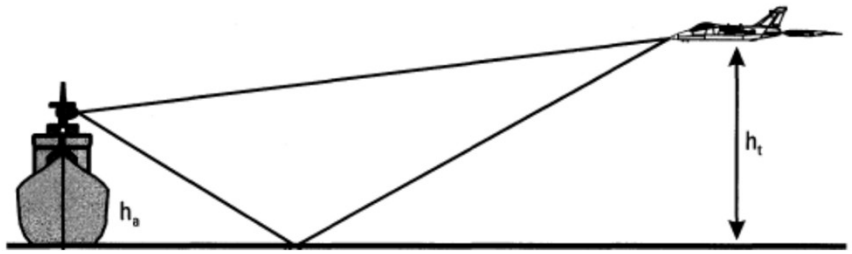
Sea State SS	Wind State $K_B = SS + 1$	Wave height (rms) σ_h
1	2	0.003m
3	4	0.2m
5	6	0.7m
7	8	1.7m

Lobing [Neri]

Smooth conducting surfaces behave like a mirror=> image of the target,

- direct and reflect ray combination,
- variation in the range assessments



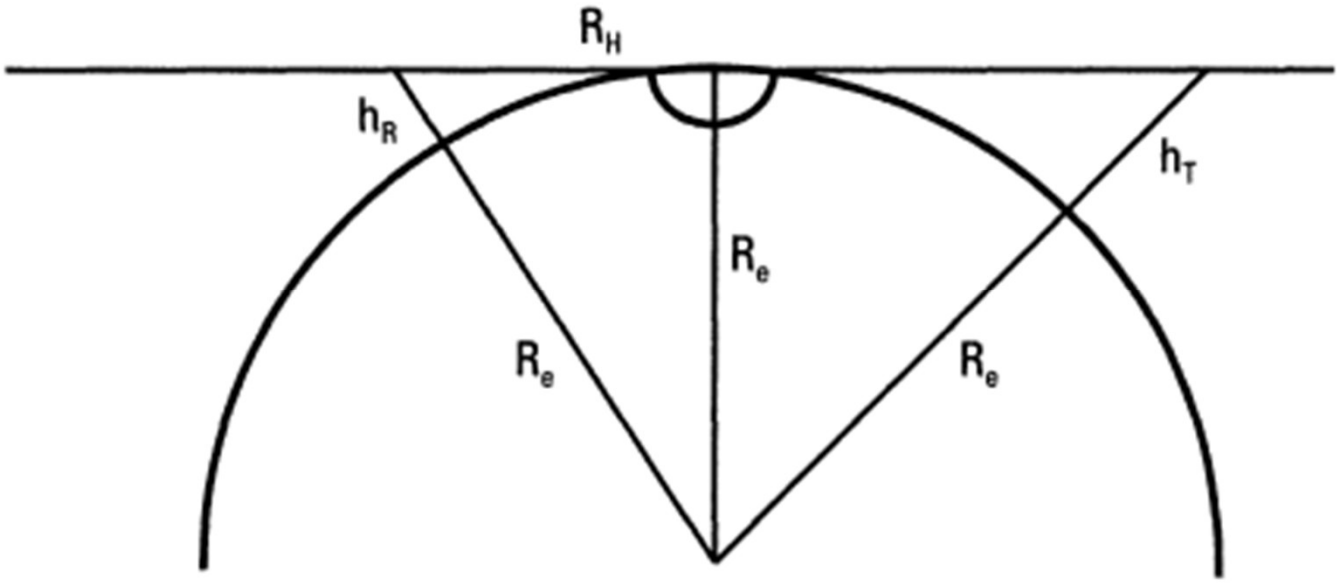


$$F_p = \left| \frac{E_d e^{jwt}\left(1 - \rho e^{j\varphi}\right)}{E_d e^{jwt}} \right| = \left|\left(1 - \rho e^{j\varphi}\right)\right|$$

Sum of two cos() with different phase and amplitudes

Sect. 8.2 [Skolnik]

Radar Horizon [Neri]



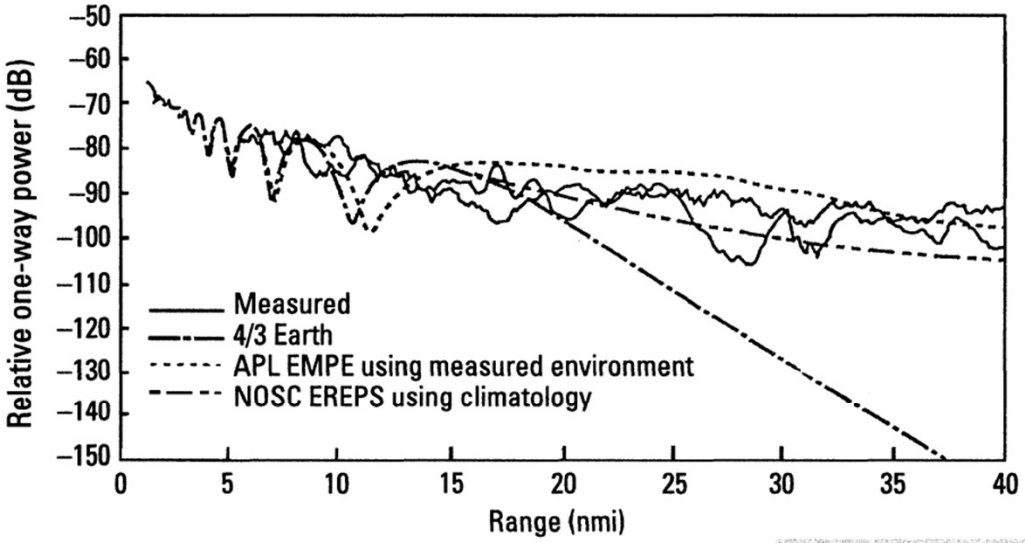
$$R_H = \sqrt{(R_e + H_R)^2 - R_e^2} \cong \sqrt{2H_R R_e} \cong 4{,}123 \times \sqrt{H_R}$$

- Additional attenuation to $1/R^2$ rate
- Assumed to be a **diffraction loss (smooth or rough surface scattering!)**

Ducting [Neri]

In some particular atmospheric conditions, and in maritime radars

- *Evaporation duct*: evaporation of moisture from the sea
- *Surface duct*: between air and sea due to discontinuity in temperature or humidity
- *Elevated duct*: at higher altitude, due to persistent air stream of particular temperature and humidity, in certain seasons and regions

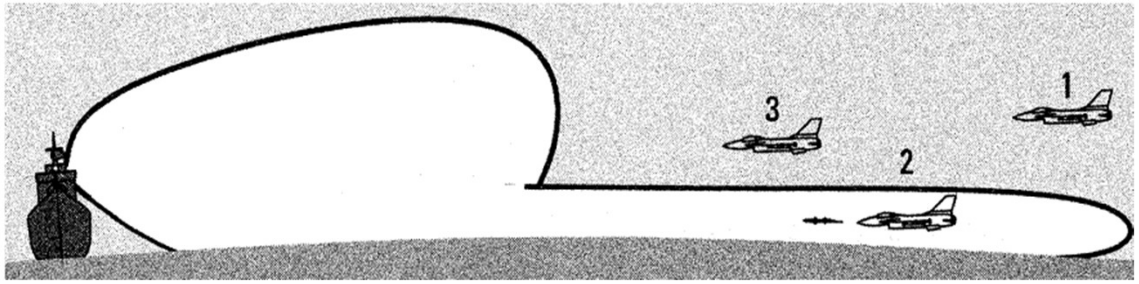


$$free\ space\ attenuation \propto 1/R^2$$

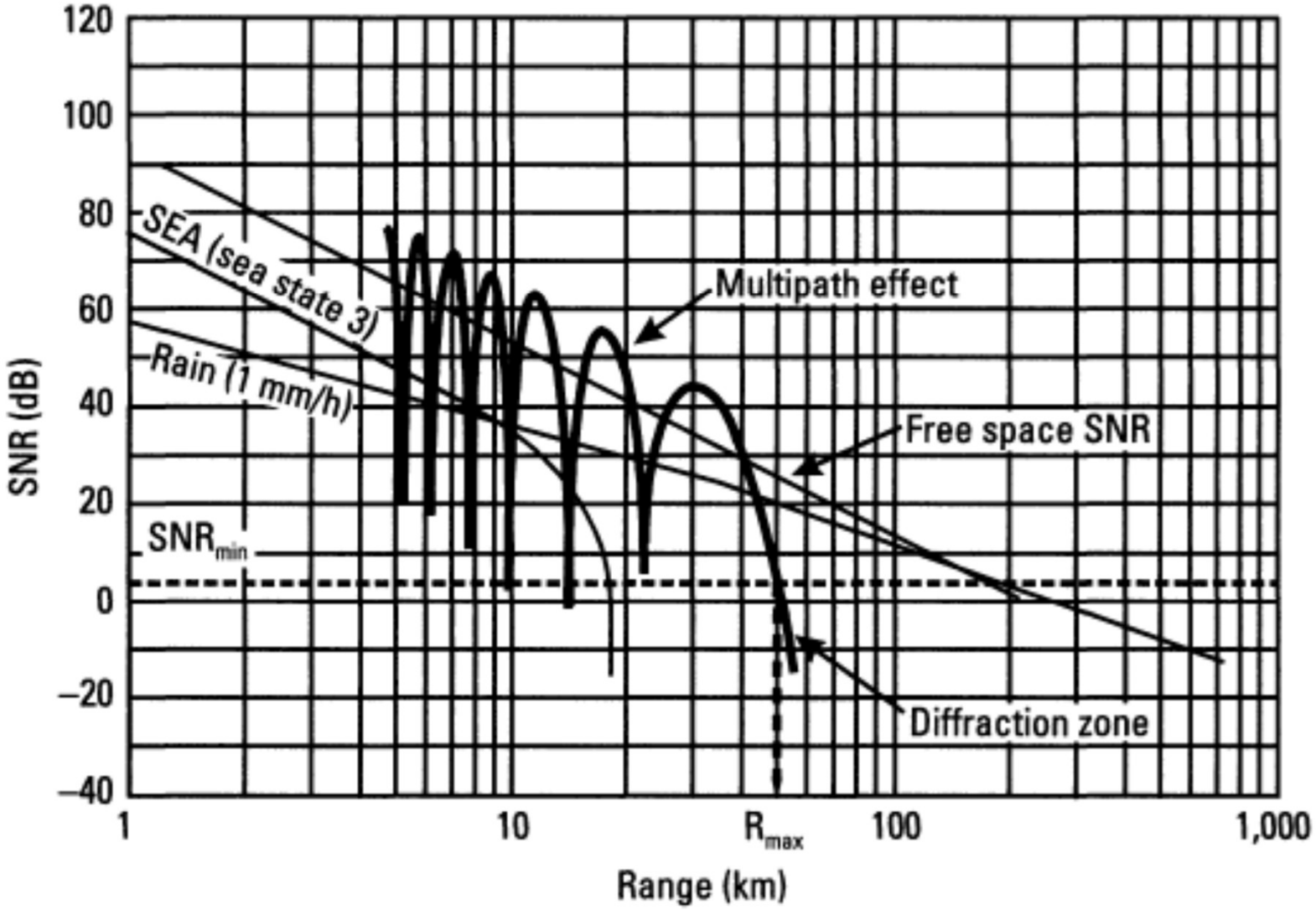
$$ducting => 1/R^{\alpha+1}$$

$$\alpha: [0,1]$$

$$\alpha = 0: trapped\ waves$$



Channel: Radar Range in operational environment



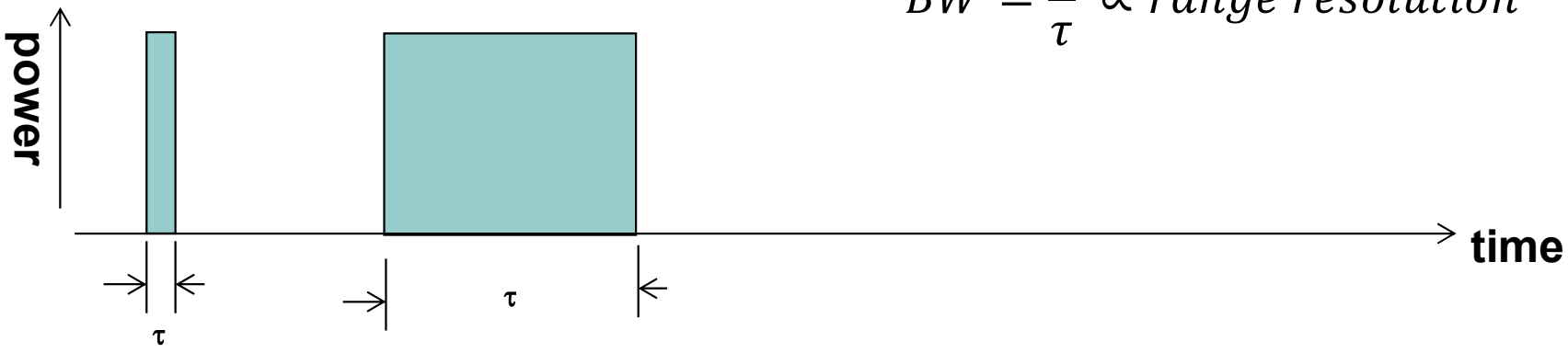
Combining all effects

Long range and finer range resolution??

- Long pulse -> longer range (longer pulse ≡ higher pulse energy)
  - Smaller bandwidth but less interference, less complexity
- Short pulse -> finer range "resolution"
  - Higher bandwidth, more interference, low dynamic range, low SNR, more complexity
- Long range+high range resolution->Short pulse with high peak power
  - High voltage -> voltage breakdown (arc discharge)
  - Safety issues, reliability problems
  - Size, volume, cost etc.

$$E = P\,\tau \;\propto range$$

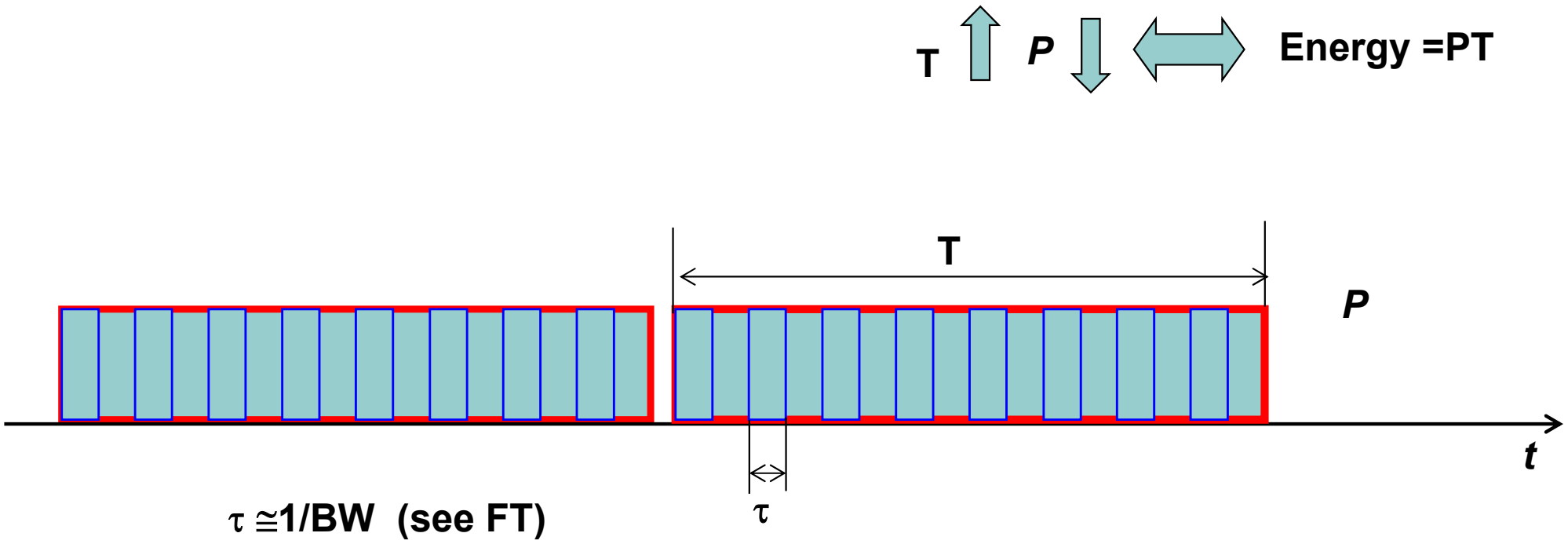
$$BW = \frac{1}{\tau} \propto range\ resolution$$



Solution: pulse compression

a long pulse consisting of many short pulses

- Higher (finer) range resolution
- Longer range



Example

Classical approach

Required range resolution: 0,3 (30 cm)

Required BW: $c\tau/2=0,3m \Rightarrow \tau = 20us \Rightarrow 1/\tau \cong B = 500$ MHz

Required Pulse energy: 1mJ

Required pulse power=1mJ/20us=**50W**

Pulse compression

Required range resolution: 0,3m (30 cm)

Required Pulse duration (long)=1 ms

Compression rate: T/τ=1ms/20us=50

Required pulse Power=1mJ/1ms=**1W**

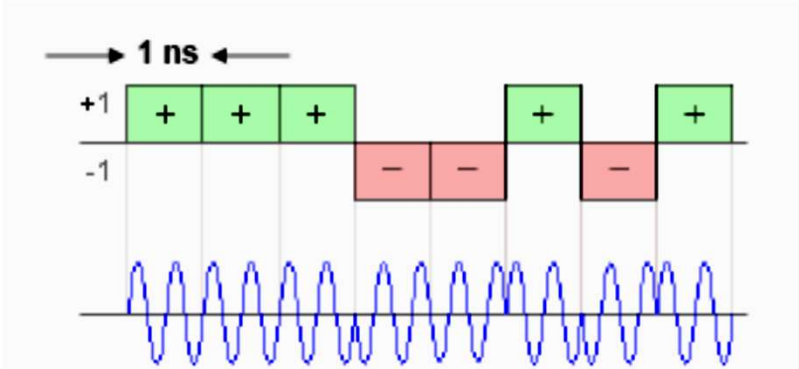
Coded radar

Pulse compression radar or coded radar
Use Modulation On Pulse (MOP),

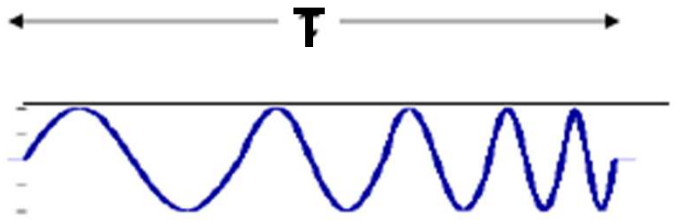
- Phase MOP (PMOP)
- Frequency MOP (FMOP)

Examples

- Low Probability of Intercept/detection (LPI/LPD) radars
- Obtain optimum clutter cancellation,
- Low peak power applications of airborne systems



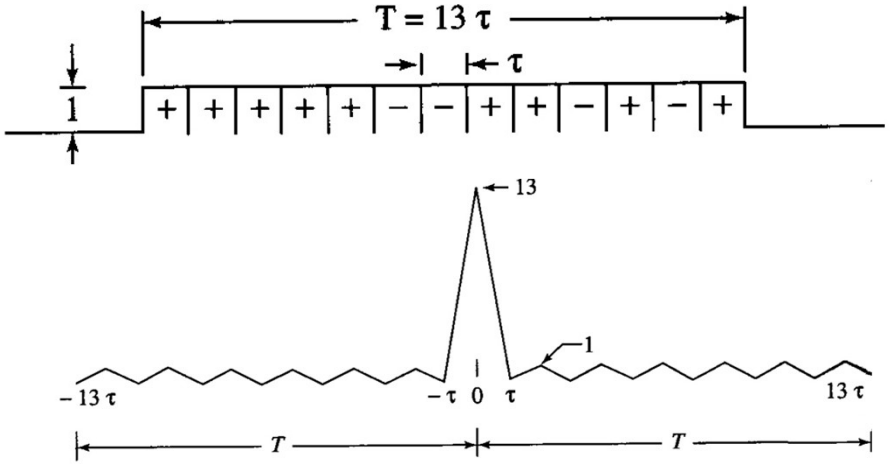
Phase coded pulses (PMOP)



Frequency modulated pulses (FMOP)

PMOP and FMOP [Neri, Skolnik]

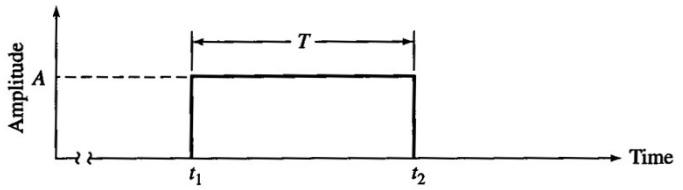
- Change the frequency (FMOP) or phase (PMOP) of carrier in each short pulse duration.
- For a given range and range resolution cell, a pulse compression radar with peak power P and n coded elements is the same as a uncoded radar with peak power nP and pulse duration equal to the duration of the code element τ



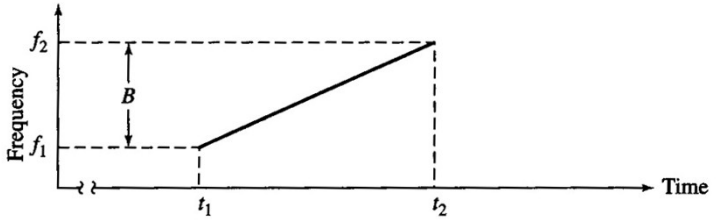
Barker Code: use a criterion in order to have equal amplitude sidelobes with SLL=-22.3 dB

Rx-> autocorrelation (matched filter)

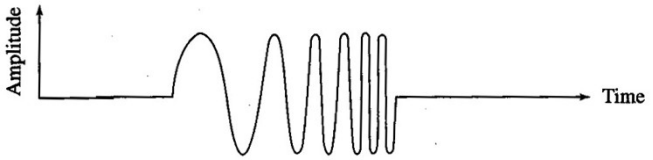
pulse



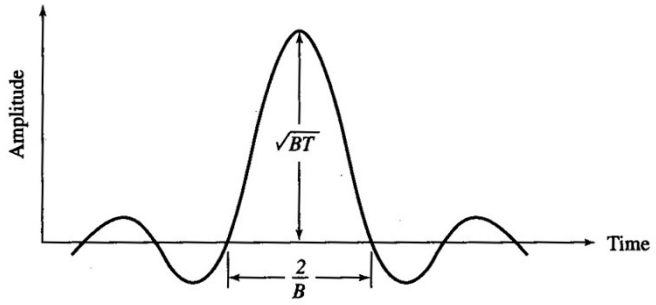
Up-chirp



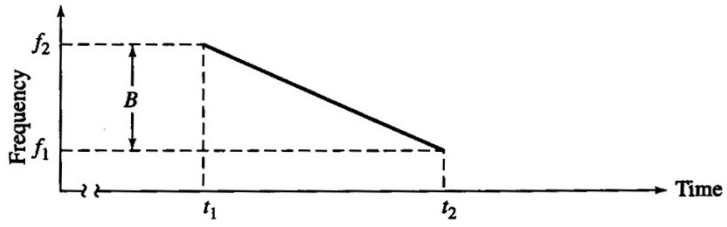
waveform



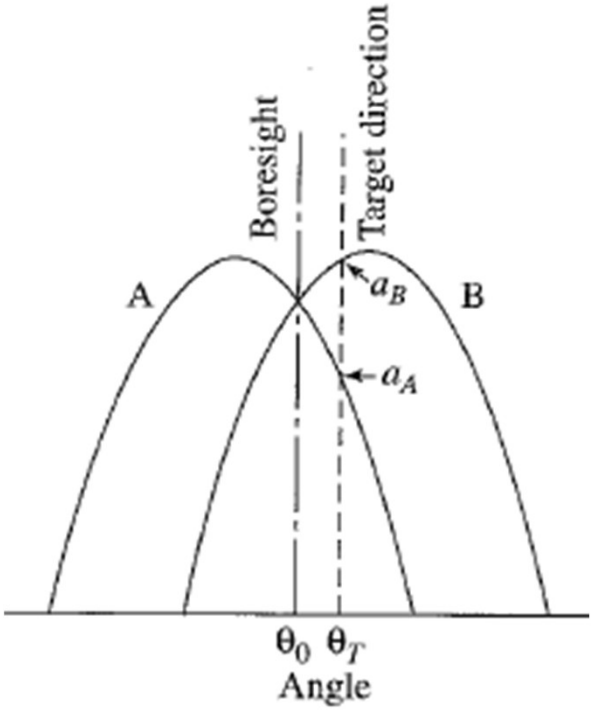
Rx compressed



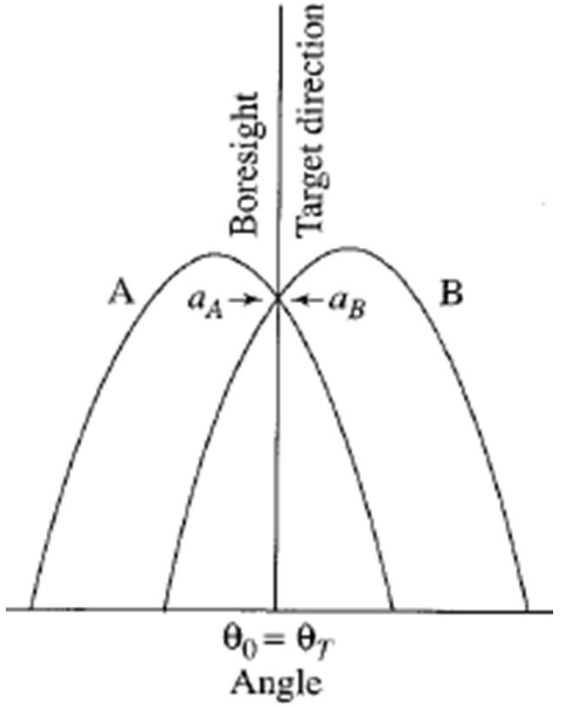
down chirp



Angle tracking



- Target echo from B is greater than the echo from A

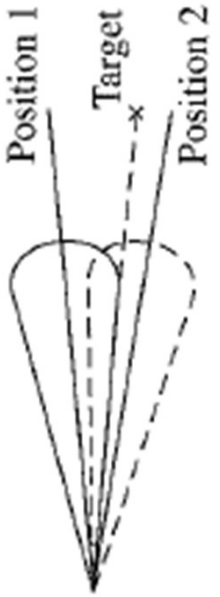


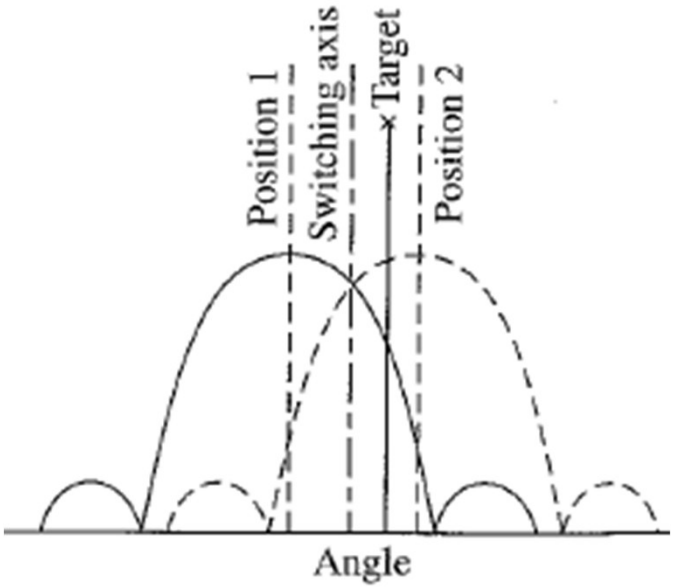
- Echo signals from B and A are equal

Two beams are squinted (Θ_q)
Crossover of two beams determines the boresight
Simultaneous beam generation or switching between several beams ?

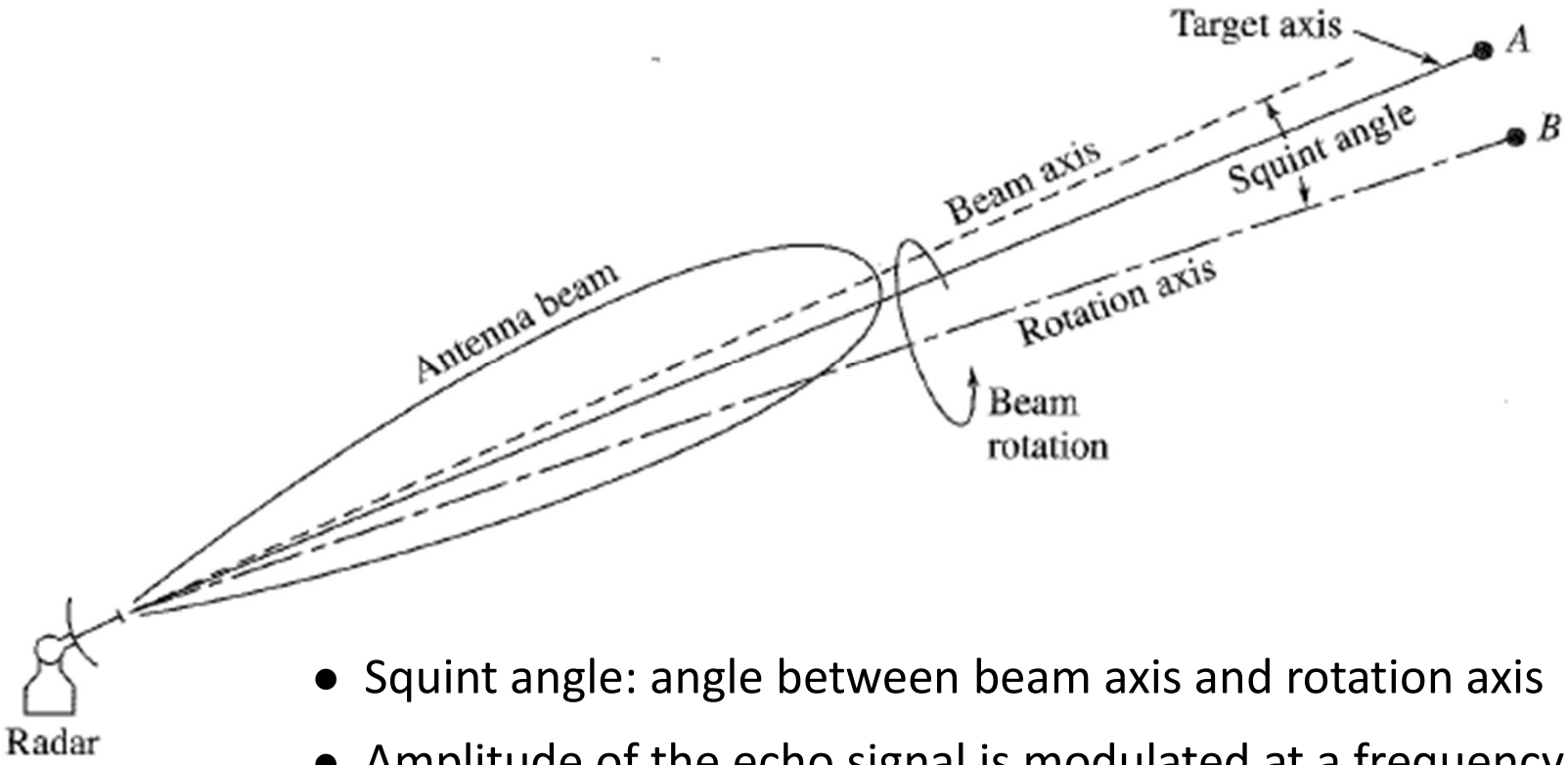
Tracking Types: Sequential lobing

- Use single antenna beam, in time sharing basis
- Simple but less accurate
- Switching of the beam between two squinted angular position
- Alternatively, called
  - •Lobe switching
  - •Sequential switching
  - •Sequential lobing
- Signal amplitude difference of two beams giving the position shift of the target from the switching axis





Tracking Types:Conical Scan (ConScan)

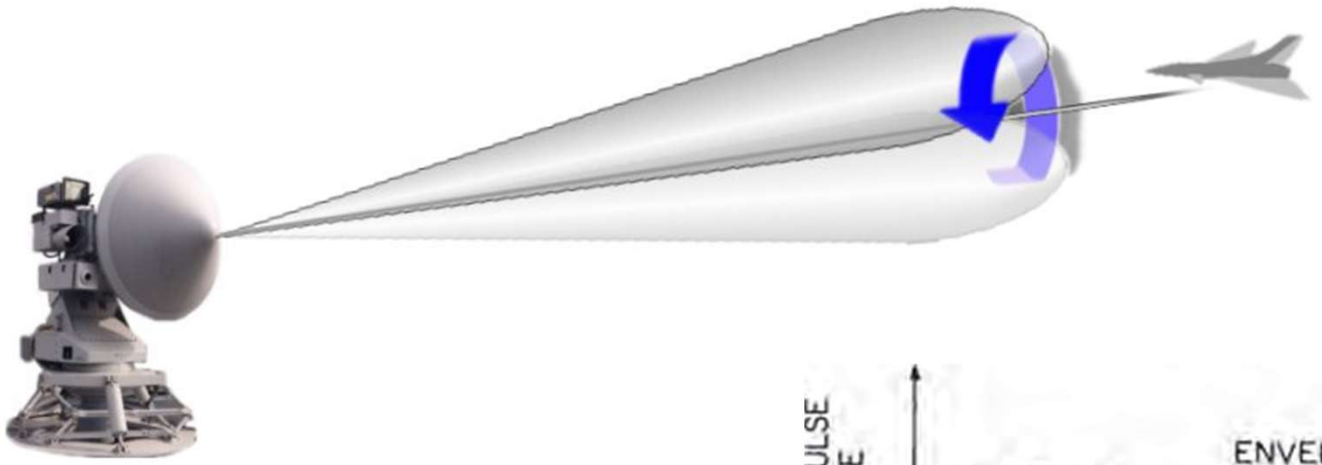


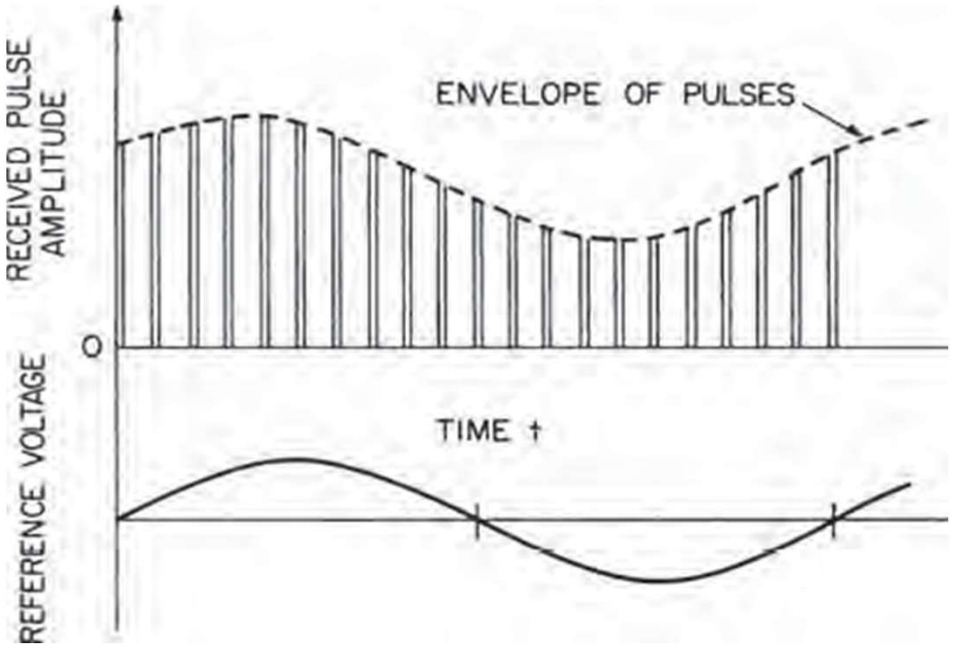


Land roll, J band
35 km: acquisition
30 km: tracking
25 km: guidance

- Squint angle: angle between beam axis and rotation axis
- Amplitude of the echo signal is modulated at a frequency of beam rotation (conical scan frequency)
- Amplitude modulation depends on angular distance between rotation axis and target location

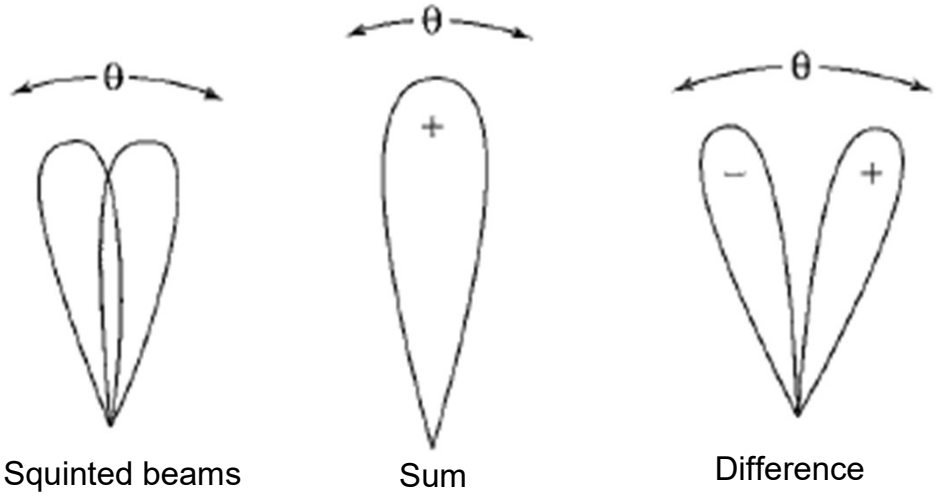
Conscan: Envelope of pulses





Tracking Types: Monopulse

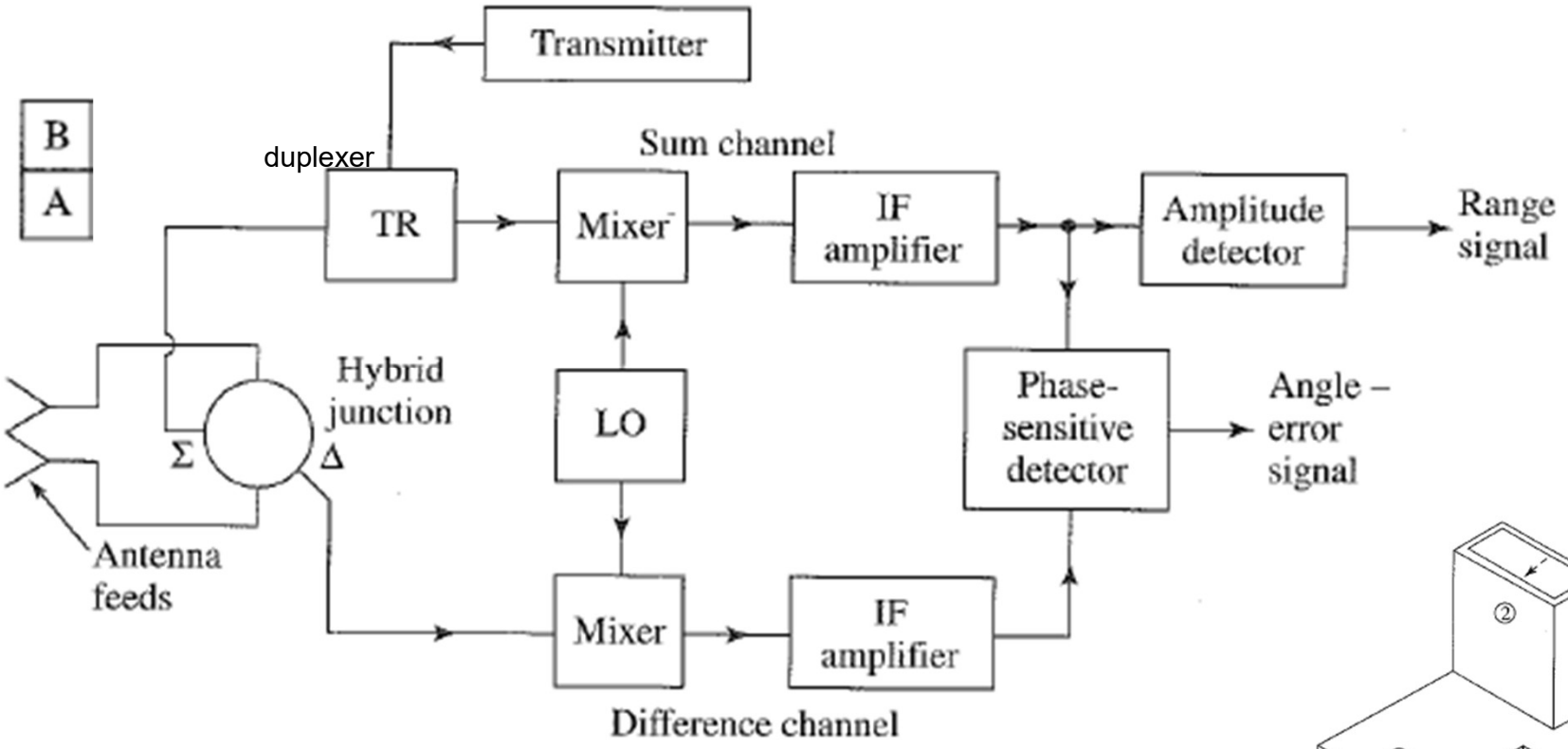
- Angular location is based on information from two (or four) simultaneous beams
- Measurement based on single (MONO) pulse, multiple pulses for higher detection probability
- Sum is used in transmission, sum and difference are used in reception
  - Amplitude comparison
  - Phase comparison



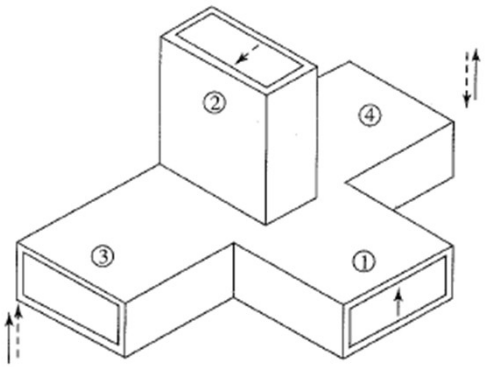


- AN/MPS-39 C-band phased array monopulse
- Multi-Object Tracking Radar-MOTR
- Missile range instrumentation radar

Monopulse radar: Amplitude comparison (el)

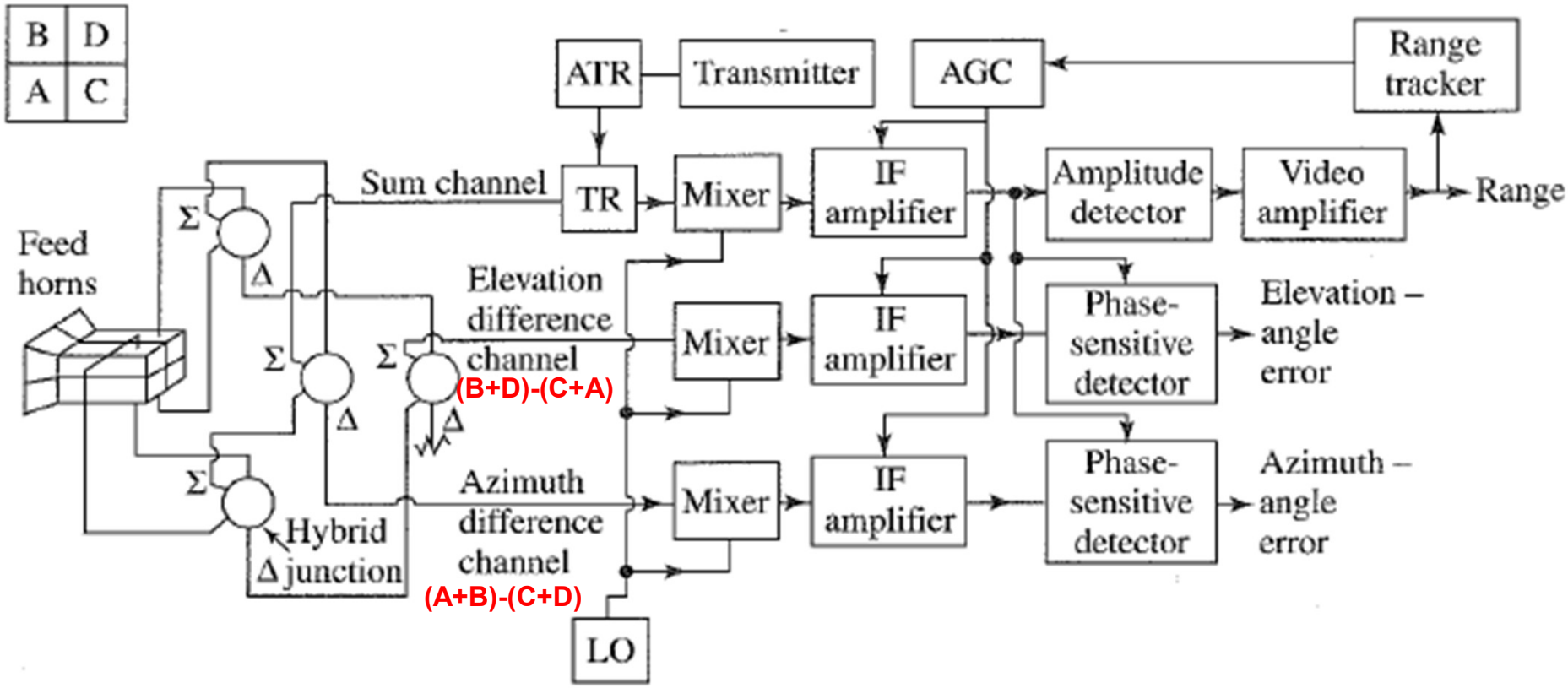


Single plane tracking (el)



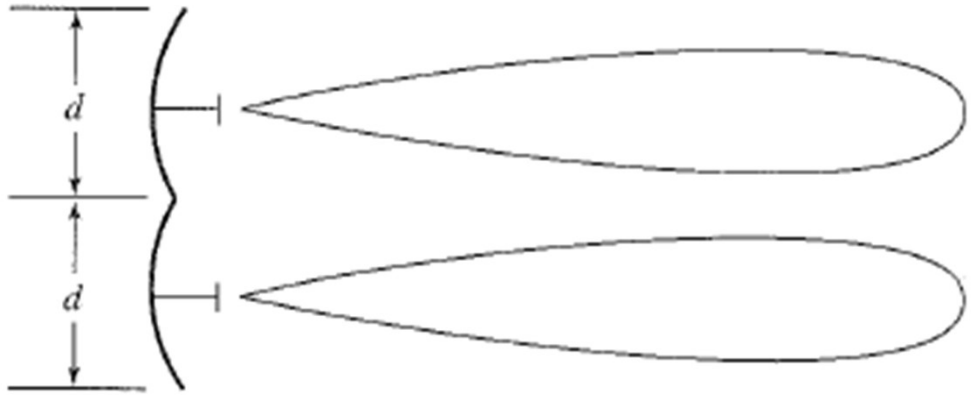
Hybrid junction (magic T)

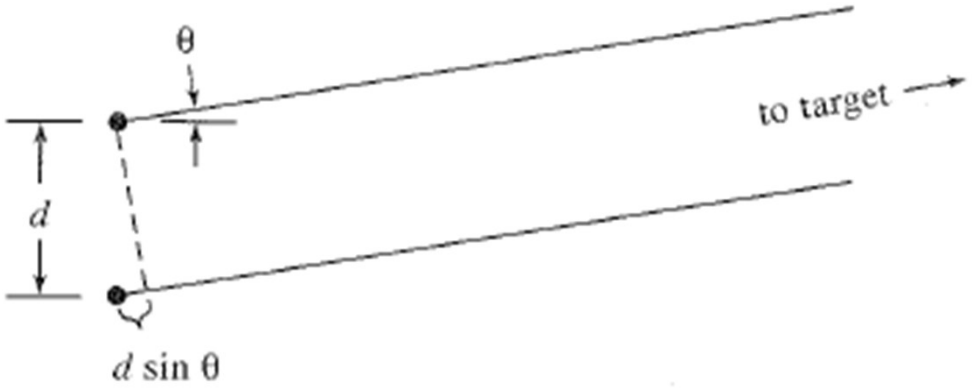
Monopulse radar in az/el: amplitude comparison



Monopulse: Phase comparison

- Two beam look at the same space rather than a squint angle
- Use two antennas to look at the same direction (rather than two feeds!)
- Amplitude of the received signals are the same but phases are different
- Just the opposite of amplitude comparison
- Sometimes called "Interferometer radar"

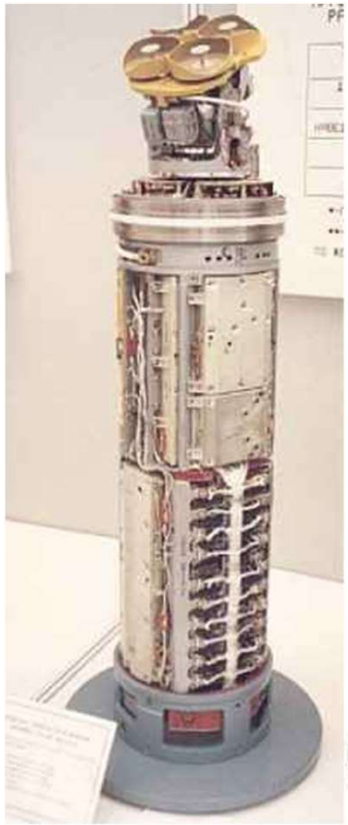




Phase difference

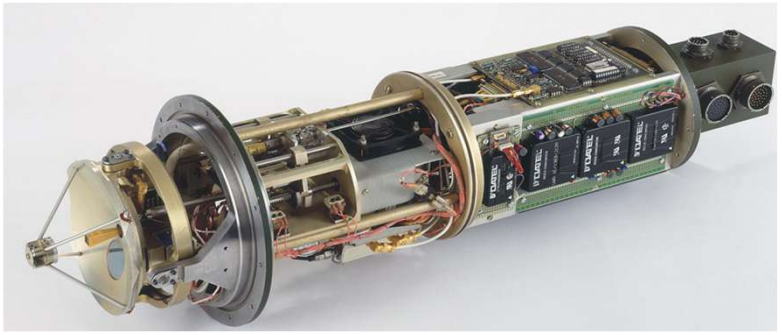
$$\Delta\phi = 2\pi\frac{d}{\lambda}\sin\theta$$

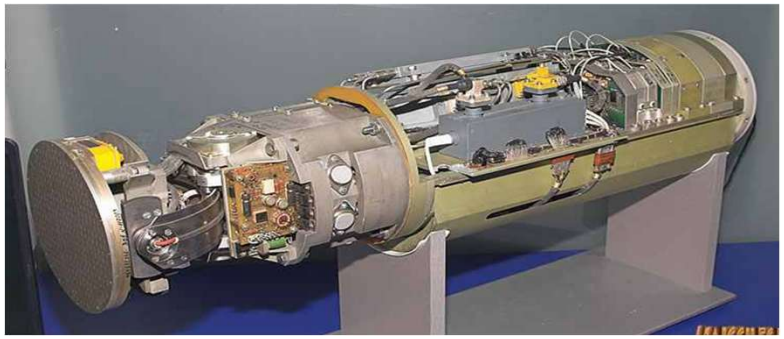
Missile seeker (tracker) [Internet]



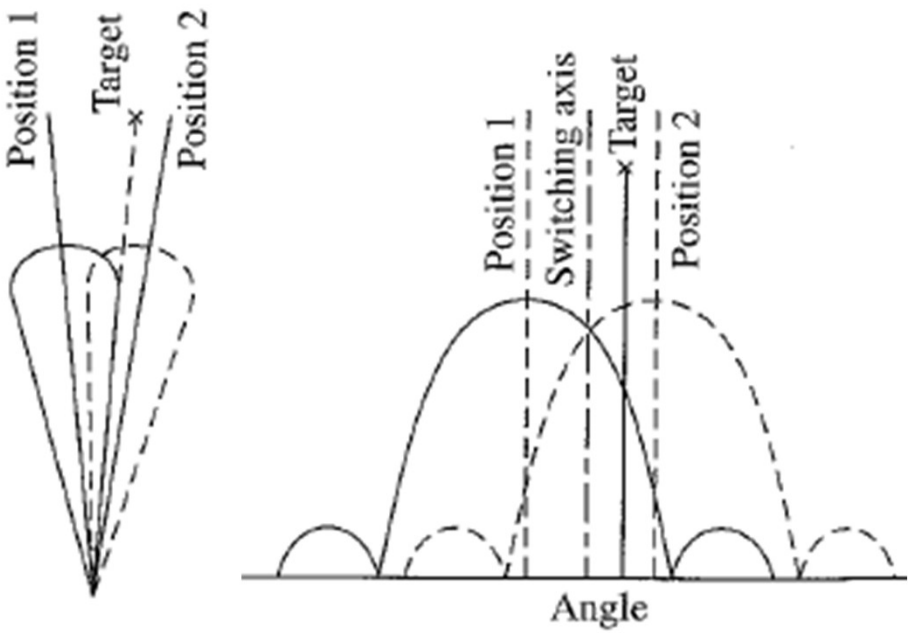


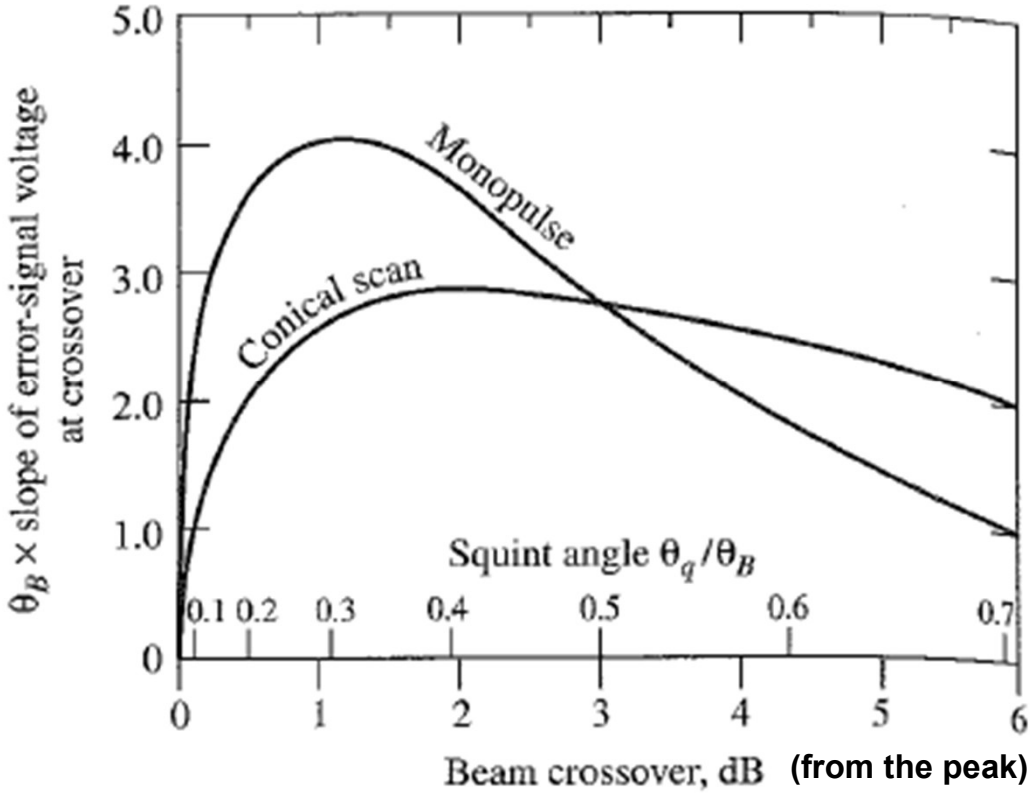






Optimum Squint angle





- The greater the squint angle,

  the greater the slope of the error (difference) signal, resulting more accurate measurement but smaller the gain of the antenna at boresight

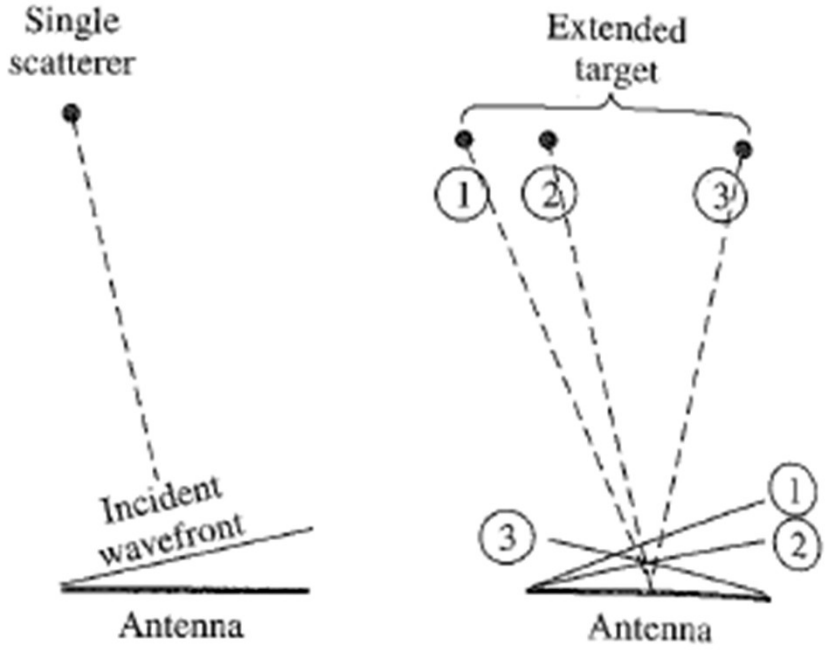
- Optimum value should be determined (optimum~ 0.31)

Accuracy of trackers: sources of error

- Glint or angle noise: affects all tracking radars, especially at short ranges
- Rx noise: affects all radars, especially at long ranges
- Amplitude fluctuations: affectting CONSCAN and sequential lobing but not monopulse (constant error)
- Servo noise: mechanical instability of servomechanism (constant error)

Glint (angle noise)

- Glint= Angle noise= angle fluctuations= angle scintillation
- Target made up of many scattering centers
- Reflected echoes independently from the centers, arriving slightly different angles at the Rx
- Vectorial sum of the echoes, resulted in a **non-uniform phase front at the planar aperture**
- Causing an error in angle measurement, a major source of error in angle measurement
- Especially at short ranges, angular extent of the target be large
- Glint mostly deterministic as long as the scattering properties of target is known



Error in angle measurements

Error in angle measurement

$$\delta_{ang} = \frac{k\theta_B}{k_s\sqrt{B\tau(S/N)\left(f_p/\beta_n\right)}}$$

$k = 1$ for monopulse and $k = 1.4\ f$ or conical scan

θ_B: half power beamwidth of the antenna

k_s: slope of the angle−error signal at boresight (different for conscan and monopulse)

B, τ: bandwidth and pulse width

f_p, β_n: PRF and servo bandwidth

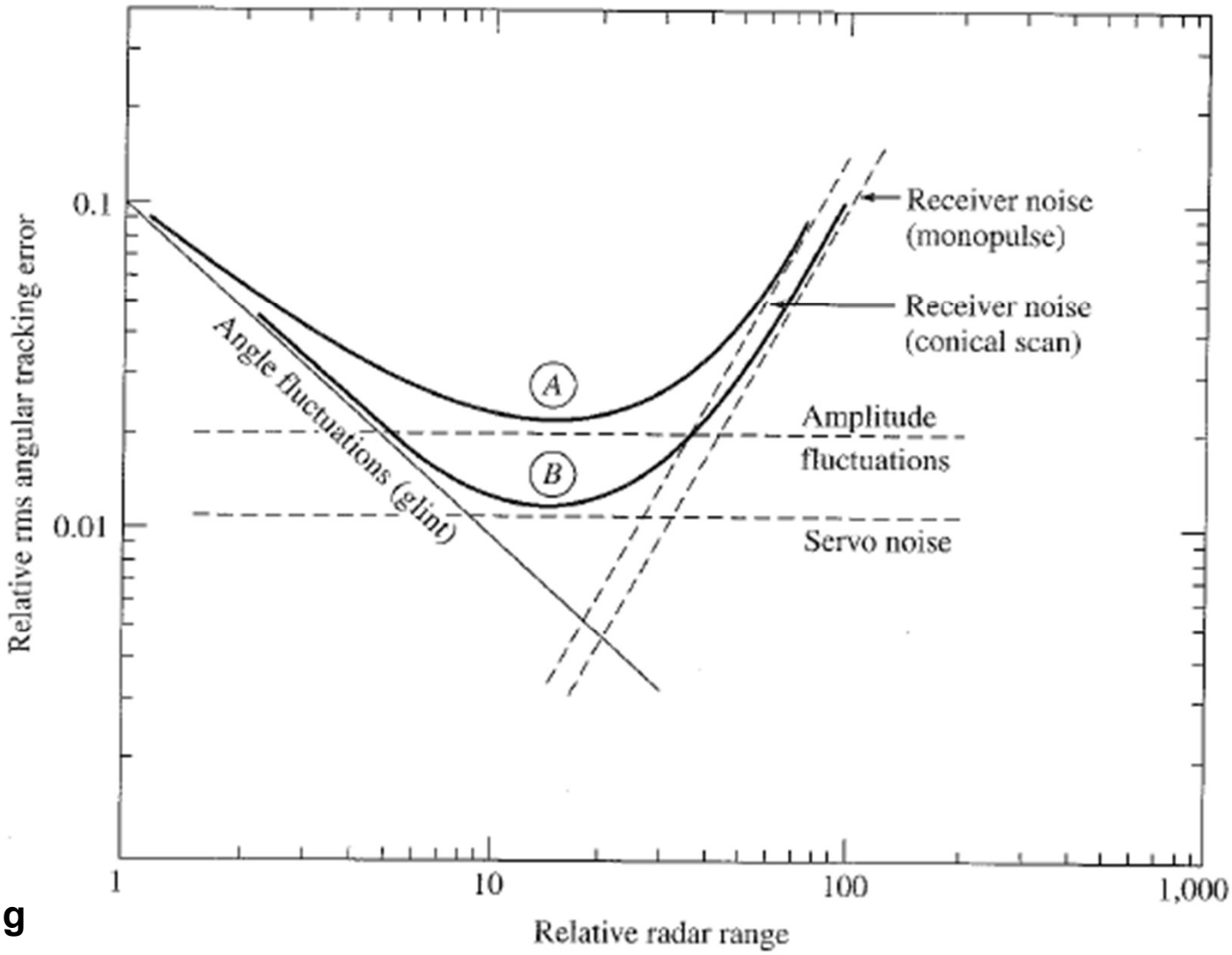
Angle tracking error (combined)

Relative contributions to the angle tracking error

- glint,
- RX noise
- Amp. Fluctuations
- Servo noise

Servo noise:

Mechanical imperfections on the system including antenna mounting structures



A: ConScan&sequential lobing
B: monopulse
Best accuracy achieved : 0.1 mrad (1 mil= 1 mrad)